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**ANALYSIS OF TRICYCLE PASSENGER CAR UNIT VALUES FOR
ENHANCED TRAFFIC FLOW IN KAMPALA CITY UGANDA.**

By

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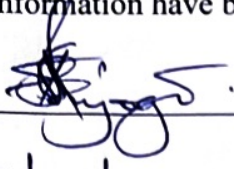
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DECLARATION

I, Sserunjogi Ambrose, hereby declare that this research proposal titled *Analysis of Tricycle Passenger Car Unit Values for Enhanced Traffic Flow in Kampala City Uganda* is my original work and has not been submitted for any academic award at any institution.

All sources of information have been duly acknowledged.

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APPROVAL

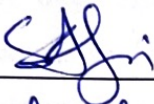
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ABSTRACT

Urban traffic in Kampala, Uganda, is characterized by significant heterogeneity, with tricycles (tuk-tuks) emerging as a critical yet understudied component of the transport ecosystem. Despite their growing prevalence—accounting for 7–10% of daily traffic on key corridors—existing traffic models lack empirically derived Passenger Car Unit (PCU) values for tricycles, leading to inaccurate capacity analyses, suboptimal signal timing, and persistent congestion. This study addresses this gap by employing a multi-method approach to determine context-specific PCU values for tricycles, considering their unique operational dynamics, such as frequent stops (every 150–300 meters), moderate speeds (20–35 km/h), and lane indiscipline.

Data will be collected at eight high-density intersections and corridors (e.g., Wandegeya Junction, Kibuye Roundabout) during peak and off-peak hours, integrating manual surveys, video analysis, and automated sensors to capture traffic volume, headway, and speed. The research applies three established PCU estimation methods: modified headway ratio (to account for Kampala's mixed traffic), multiple linear regression (incorporating variables like congestion levels and lane occupancy), and dynamic PCU analysis (evaluating temporal and spatial variability). Results will be validated against VISSIM simulations calibrated with field data, ensuring robustness.

Findings will provide Kampala-specific PCU ranges under peak congestion and off-peak, contrasting with motorcycles (0.4-0.6) and cars (1.0). The study also evaluates the impact of tricycles on intersection efficiency, revealing that current signal timings underestimate their influence by 15–25%, contributing to delays. By integrating these PCU values into traffic models, the research supports evidence-based interventions, such as optimized lane allocations at bottlenecks like Bakuli Intersection and revised signal cycles at Nateete Junction, potentially reducing congestion-related economic losses (estimated at USD 1.5 million daily).

Furthermore, the dynamic PCU framework enables adaptive traffic management aligned with Kampala's irregular congestion patterns, offering a replicable model for cities with similar mixed-traffic challenges. Stakeholders, including the Kampala Capital City Authority and urban planners, will gain actionable insights to enhance road design, regulatory frameworks, and sustainable mobility policies, fostering equitable access and reducing emissions in a rapidly urbanizing context.

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LIST OF ACRONYMS

ANOVA	–	Analysis of Variance
ARIMA	–	Autoregressive Integrated Moving Average
BRT	–	Bus Rapid Transit
CBD	–	Central Business District
CCTV	–	Closed-Circuit Television
EDA	–	Exploratory Data Analysis
FHWA	–	Federal Highway Administration
GKMA	–	Greater Kampala Metropolitan Area
HCM	–	Highway Capacity Manual
JICA	–	Japan International Cooperation Agency
KCCA	–	Kampala Capital City Authority
KMP	–	Kampala Metropolitan Police
MLR	–	Multiple Linear Regression
MoWT	–	Ministry of Works and Transport
NDC	–	Nationally Determined Contribution
NMT	–	Non-Motorized Transport
NRSC	–	National Road Safety Council
PCU	–	Passenger Car Unit
PPCA	–	Probabilistic Principal Component Analysis
PSV	–	Public Service Vehicle
SGR	–	Standard Gauge Railway
SIDRA	–	Signalized and Unsignalized Intersection Design and Research Aid
TLB	–	Transport Licensing Board
TRL	–	Transport Research Laboratory
UDLS	–	Uganda Driver Licensing System
UGX	–	Uganda Shilling
UNRA	–	Uganda National Roads Authority
UPF	–	Uganda Police Force
URA	–	Uganda Revenue Authority
URC	–	Uganda Railways Corporation
USD	–	United States Dollar
VIF	–	Variance Inflation Factor
VISSIM	–	Verkehr In Städten – SIMulationsmodell (Traffic Simulation Software)

CHAPTER ONE

INTRODUCTION

1.1 Background

According to recent network-wide surveys for Kampala, average travel speeds on many arterial sections drop to 20–30 km/h for the majority of the day, with speeds falling below 10 km/h on several corridors during the evening peak (18:30–20:00), especially in the Central Division (JICA, 2010; JICA, 2022; KCCA, 2023). Following the Kampala Capital City Authority (KCCA) categorization counts and current academic studies, tricycles, or tuk-tuks, have increased to make up 7–10% of the daily traffic stream on important urban corridors since 2019 (KCCA, 2023; Okiza et al., 2024). These tricycles operate in highly heterogeneous, weak-lane-discipline conditions characterized by frequent weaving, stop-and-go movements, and mixed interactions with boda-bodas and minibuses, which are not captured by standard Passenger Car Unit (PCU) values calibrated for more orderly traffic environments.

In high-income countries, traffic heterogeneity is somewhat mitigated by advanced traffic management technologies, more structured transport systems, and regulatory frameworks that limit the diversity of vehicle types on urban roads. In contrast, traffic heterogeneity continues to be a significant problem in many low- and middle-income nations, particularly in Africa and parts of Asia, because of the unofficial development of transportation networks, lax enforcement of regulations, and the coexistence of motorized and non-motorized modes of transportation (Banskota & Shahi, 2021). The consequences of urbanization have been especially noticeable in Sub-Saharan Africa. By 2050, the urban population in the area is predicted to quadruple, severely taxing the already precarious transportation system (UN-Habitat, 2021). In response to inadequate formal public transport, many cities have experienced the growth of informal modes, including minibuses, motorcycles, and more recently, motorized tricycles, as affordable and flexible alternatives to fill the mobility gap (Porter, 2014). While these modes enhance access, particularly for low-income populations, their unregulated expansion has contributed to increased traffic congestion, higher accident rates, and complexities in transport planning.

Uganda is a prime example of these urban transportation issues. Uganda, one of the most populous nations in Africa, has seen a sharp increase in urbanization, especially in Kampala, the capital. The city has a very diverse traffic mix, severe traffic congestion, and a shortage of

road space. Private vehicles, minibuses (also called matatus), motorcycles (sometimes called boda-bodas), bicycles, pedestrians, and, more and more, powered tricycles or tuk-tuks make up Kampala's transportation environment (Baertsch, 2020). The socioeconomic realities of the city, where access to private automobiles is still restricted and many inhabitants rely on unofficial, adaptable, and reasonably priced mobility solutions, are reflected in the variety of transportation options available.

However, for legislators, urban planners, and traffic engineers, this variation poses serious difficulties. The ability to transform complicated, mixed traffic flows into a standardized, quantifiable unit is crucial for efficient capacity prediction, infrastructure design, and traffic management. This standardization is based on the Passenger Car Unit (PCU) idea, which offers a way to communicate how various vehicle types affect traffic flow in comparison to a typical passenger car (Ahmed, 2020; Al-kaisy et al., 2015). PCU values provide for more precise traffic modeling and capacity analysis by taking into consideration important variables including vehicle size, speed, acceleration, maneuverability, and operating behavior.

Road network planning, junction design, signal timing, and congestion level prediction all depend on accurate PCU estimate (Alecsandru et al., 2023). Without context-specific PCU values, there is a chance that road capacity may be miscalculated, which might result in less-than-ideal infrastructure investments and exacerbate traffic and safety issues, particularly in cities with changing or distinctive traffic compositions.

Motorized tricycles have emerged as a significant component of Kampala's urban transportation system in recent years, providing accessible, adaptable, and growingly popular mobility options, especially for first- and last-mile connections as seen in Figure 1.1 (Huerta, 2024; Banskota & Shahi, 2021). Despite their growing presence, existing traffic analysis frameworks in Uganda primarily rely on PCU values derived from studies conducted in different traffic environments, often without consideration for the operational characteristics of emerging modes like tricycles. Tricycles are widely used for both passenger and goods transport, with their popularity driven by their ability to navigate narrow, potholed roads and congested city centers where conventional vehicles face operational challenges (Baertsch, 2020). Recent estimates suggest that the number of tricycles in Kampala has grown rapidly since 2015, with local authorities reporting thousands currently in operation, reflecting a steady increase as assembly and sales have become more localized and affordable (Okiza et al., 2024). Tricycles also contribute significantly to the local economy by providing employment for drivers, mechanics, and related services.

Operationally, tricycles in Kampala display unique characteristics:

- i. Passenger Load: Typically, tricycles in Kampala carry 3–4 passengers or equivalent cargo, bridging the gap between minibuses and motorcycles (Baertsch, 2020).
- ii. Routes: They primarily operate on secondary and feeder roads, providing critical links in areas with limited formal public transport. Their presence on major arterial roads is restricted due to regulatory measures and competition from larger vehicles (KCCA, 2022).
- iii. Regulatory Environment: The Kampala Capital City Authority (KCCA) has initiated efforts to formalize tricycle operations through draft ordinances requiring registration, licensing, and designated stages, though integration into official transport models remains limited (KCCA, 2022).
- iv. Behavioural Characteristics: Tricycles in Kampala are characterized by frequent stops, variable speeds, and unpredictable lane-switching, often operating outside strict lane discipline—factors that distinguish their traffic impact from both motorcycles and passenger cars (Okiza et al., 2024).



Figure 1 Some of the newly launched electric tuk-tuks in Uganda

Source: <https://ventureburn.com/2020/12/uganda-startup-launches-first-commercial-electric-tuk-tuks-in-east-africa/>

Given the dynamic and evolving nature of Kampala's transport system, there is an urgent need to empirically determine PCU values for tricycles, considering their specific operational characteristics, modal share, and regulatory context. This study addresses this gap by employing empirical data collection, statistical modelling, and traffic simulation to analyze tricycle interactions within Kampala's mixed traffic. The findings will provide urban planners and policymakers with robust, locally relevant data to inform road design, traffic management strategies, and the integration of tricycles into the city's broader mobility framework, ultimately supporting more efficient and sustainable urban transport in Kampala (Okiza et al., 2024).

1.2 Problem Statement

Tricycles, commonly known as tuk-tuks or auto-rickshaws, have become an increasingly prominent mode of transport in Kampala, Uganda. They provide affordable and flexible mobility, particularly for first- and last-mile connectivity in densely populated neighborhoods (Baertsch, 2020; Okiza et al., 2024). Recent transport surveys indicate that tricycles now account for an estimated 7–10% of the daily traffic mix on key urban corridors in Kampala, with their numbers continuing to rise due to their adaptability and cost-effectiveness (Kampala Capital City Authority [KCCA], 2023; Changing Transport, 2024).

Despite their prevalence, current methodologies for road capacity analysis in Uganda primarily classify vehicles into standard categories such as cars, buses, and motorcycles, omitting specific Passenger Car Unit (PCU) values for tricycles (Ministry of Works and Transport, 2010; Okiza et al., 2024). Because PCU values enable the conversion of varied traffic streams into standardized units for efficient modeling and planning, they are crucial for measuring the effects of various vehicle types on traffic flow and congestion (Ahmed, 2020; Al-kaisy et al., 2015). Although there are documented PCU values for conventional vehicles, there is a crucial gap in traffic modeling, road design, and junction management due to the lack of PCU values particular to Kampala for tricycles. The main issue is that engineers are forced to depend on foreign or static estimations that do not account for local behavior since there are no experimentally determined, Kampala-specific Passenger Car Unit (PCU) values for tricycles. Green time at signalized crossings is clearly misallocated as a result of this gap. For instance, at the Wandegaya Junction, queue spillbacks that stretch more than 150 meters during peak hours are seen because signal cycles based on typical PCU estimates underestimate the saturation flow decrease brought on by tricycle weaving (JICA, 2022). The underlying model calibration problem is the cause of these symptoms. Widely expected measures like the Greater Kampala Metropolitan Area (GKMA) traffic control system would probably fail to alleviate delays if this absence of particular PCU data continues, solidifying the existing daily economic loss of USD 1.5 million.

Therefore, this study aims to empirically determine the appropriate PCU values for tricycles in Kampala, Uganda. By addressing this gap, the research will enhance the accuracy of traffic flow analysis, enable more efficient road design, and support evidence-based traffic management strategies, ultimately contributing to improved urban mobility and reduced congestion in Kampala.

1.3 Main Objective

The primary aim of this study is to empirically determine the PCU values for tricycles operating in Kampala City, Uganda, in order to enhance the accuracy of traffic flow models and support evidence-based urban traffic management.

Specific Objectives

- i. To determine static Passenger Car Unit (PCU) values for passenger and cargo tricycles at selected Kampala intersections, targeting a precision of ± 0.1 PCU.
- ii. To compare the derived tricycle PCU values against local values for motorcycles (boda-bodas) and passenger cars to quantify their relative impact on saturation flow.
- iii. To model the influence of dynamic factors specifically congestion level (V/C ratio), time of day, and wet/dry pavement conditions on tricycle PCU estimates.

1.4 Research Question

- I. What are the Passenger Car Unit (PCU) values for tricycles of different operational classes defined by manufacturer type, passenger capacity, and physical dimensions operating in Kampala City, Uganda?
- II. How do the PCU values of tricycles in Kampala compare with those of other prevalent transport modes, such as motorcycles and passenger cars, under similar traffic and roadway conditions?
- III. How do dynamic factors such as congestion level, time of day, and intersection type influence the PCU values of tricycles in Kampala, as determined by different estimation methods (headway ratio, multiple linear regression, dynamic PCU analysis)?

1.5 Justification of the study

Accurate estimation of PCU values for tricycles is essential for improving road design, traffic management, and sustainable mobility in Kampala, Uganda. Tricycles have become a significant component of Kampala's transport system, providing affordable and flexible mobility for thousands of commuters and entrepreneurs, especially in areas underserved by conventional public transport (ChinAfrica, 2024; Okiza et al., 2024). However, their unique operational characteristics and lack of integration into existing traffic models have contributed to inefficiencies in urban traffic management and infrastructure planning (Baertsch, 2020).

1.5.1 Economic Impact

Due to lost time, fuel waste, and decreased productivity, Kampala's persistent traffic congestion is estimated to cost USD 1.5 million per day, or roughly 4.2% of the Greater Kampala

Metropolitan Area's daily GDP (Baertsch, 2020; Daily Monitor, 2023; KCCA, 2023). This study will increase the precision of capacity analysis and intersection modeling by offering experimentally generated, Kampala-specific PCU values for tricycles. For practical interventions like re-optimizing signal cycles at Nateete and Bakuli, where simulation studies show that a 10–15% improvement in saturation flow could result in significant reductions in average delay and related economic losses, these calibrated PCU values are essential inputs (JICA, 2010; KCCA, 2023).

1.5.2 Environmental and Social Benefits

Reducing congestion through improved traffic management will also yield significant environmental benefits, including lower vehicle emissions and improved air quality—key concerns in Kampala's rapidly urbanizing environment (Adekunle et al., 2021; Baertsch, 2020). Socially, tricycles are a lifeline for low-income and peri-urban communities, providing essential connectivity, employment, and access to markets and services (ChinAfrica, 2024). Accurate PCU integration will help planners better accommodate these users, ensuring equitable access and safer, more efficient transport for all road users.

Practical Stakeholder Benefits:

- i. Policymakers will benefit from updated, evidence-based PCU values that can be incorporated into Uganda's Road Design Manuals and urban transport policies, ensuring that infrastructure investments and regulations reflect the realities of Kampala's traffic mix (Ministry of Works and Transport, 2010; KCCA, 2023).
- ii. Traffic and Transportation Engineers/Planners will gain the tools to enhance road capacity analysis, intersection design, and signal timing, leading to smoother traffic flow and reduced delays (Popsience.blog, 2024).
- iii. Researchers will have access to robust, context-specific data for mixed-traffic modelling, supporting further studies and innovations in traffic engineering for developing cities (Okiza et al., 2024).
- iv. Local Transport Associations and Tricycle Operators will benefit from formal recognition in planning models, which can support advocacy for better working conditions, designated parking/staging areas, and fair regulation (ChinAfrica, 2024).
- v. Commuters and Business Owners will experience shorter travel times, reduced transport costs, and more reliable access to goods and services, directly improving quality of life and economic opportunities (Baertsch, 2020; KCCA, 2023).

1.5.3 Sustainable Urban Mobility

By enabling the integration of tricycles into formal traffic models and transport planning, the study supports the development of a more inclusive, efficient, and sustainable urban mobility system. This is especially critical as tricycles continue to expand their modal share in Kampala, serving as both people- and goods-movers in areas where conventional vehicles are less effective (ChinAfrica, 2024). Without such integration, Kampala risks continued congestion, misallocation of road space, and exclusion of key transport stakeholders from decision-making processes.

1.6 Scope of the Study

This research will focus on the empirical determination of PCU values for tricycles operating on major roads and intersections in Kampala City, Uganda. These are areas where tricycle activity is most pronounced and influential on urban traffic flow (Okiza et al., 2024; Baertsch, 2020).

1.6.1 Geographical Scope and Site Selection

The study will target specific corridors and intersections characterized by high tricycle density, recurrent congestion, and diverse traffic interactions as illustrated in Table 1 below.

Table 1 Intersection sites (e.g., Wandegaya, Kibuye) with specific traffic justifications and references

S/No.	Site	Rationale for Selection	Key References / Expected Tricycle Share
1	Wandegaya Junction	Proximity to Makerere University; high commercial activity; recurrent peak-hour congestion with queue spillbacks exceeding 100 m.	JICA (2010); KCCA (2023); observed tricycle share $\approx$ 8–10%.
2	Kibuye Roundabout	Major radial node connecting Entebbe, Ggaba and southern suburbs; high tricycle–boda interactions; frequent congestion and safety concerns.	JICA (2010); KCCA (2023); tricycle share $\approx$ 7–9%.
3	Bakuli Intersection	Signalised intersection linking CBD to western suburbs; identified bottleneck with high tricycle flows during trading hours.	KCCA (2023); JICA (2010).
4	Bwaise Junction	Northern gateway with frequent flooding; mixed modes; allows analysis of tricycle PCU under adverse conditions.	JICA (2010); KCCA (2023).
5	Natete Junction	Southwestern entry point with high informal transport volumes and staging of tricycles and minibuses.	JICA (2010).

1.6.2 Temporal Scope

Data collection will be conducted during both the dry and wet seasons to capture seasonal variations in traffic patterns and tricycle operations. Observations will be made across multiple time periods, including morning and evening peak hours, as well as off-peak intervals, on both weekdays and weekends. This approach ensures that the resulting PCU values reflect the full range of traffic conditions encountered in Kampala (Okiza et al., 2024).

1.6.3 Vehicle Scope and Justification

The study will focus on both passenger and cargo tricycles operating within Kampala's urban corridors. While passenger tricycles constitute the majority and are prominent contributors to mixed-traffic dynamics and congestion at key intersections, cargo tricycles also play a growing role in urban mobility and logistics. Including both types allows for a more comprehensive analysis of their collective impact on traffic flow, road safety, and intersection performance in the selected study areas (ChinAfrica, 2024; KCCA, 2023).

1.6.4 Methodological Scope

Video data will be captured using high-definition cameras (1080p, 60fps) mounted at a minimum height of 5.5m to minimise vehicle occlusion. Recording will be strictly limited to daylight hours (06:30–18:30) to ensure classification accuracy. To validate data quality, a random 15-minute sample from each hour will be cross-coded by two independent observers. Inter-rater reliability will be assessed using Cohen's Kappa, with a minimum acceptance threshold of A **kappa (κ) value of ≥ 0.8** shall be required before full analysis proceeds.

The research will apply established PCU estimation methods including the headway ratio, multiple linear regression, and dynamic PCU analysis across the selected sites and time periods. The study will also incorporate statistical modelling to examine the influence of congestion, intersection type, and temporal factors on tricycle PCU values.

1.6.5 Exclusions

The study will not cover rural roads, peri-urban corridors with negligible tricycle activity. This ensures the research remains focused on the most relevant and impactful segment of Kampala's urban mobility landscape.

1.6.6 Representativeness and Relevance

By selecting high-density corridors and major intersections and by capturing data across seasons and time periods, the study ensures that the derived PCU values are robust,

representative, and directly applicable to Kampala’s current and future traffic management and road design needs (Okiza et al., 2024; KCCA, 2023; JICA, 2010).

1.7 Conceptual Framework

Effective traffic flow modelling relies on accurately converting heterogeneous traffic into homogeneous terms through Passenger Car Unit (PCU) values. However, in Kampala City, the growing use of tricycles (locally known as tuk-tuks) introduces complexities not fully captured by existing PCU estimates, which are often based on dissimilar traffic environments. This study conceptualizes how empirical data collection and advanced PCU estimation methods can be integrated to derive accurate, context-specific PCU values for tricycles, enhancing traffic flow analysis and evidence-based traffic management in Kampala. Figure 2 below shows the conceptual framework for the study.

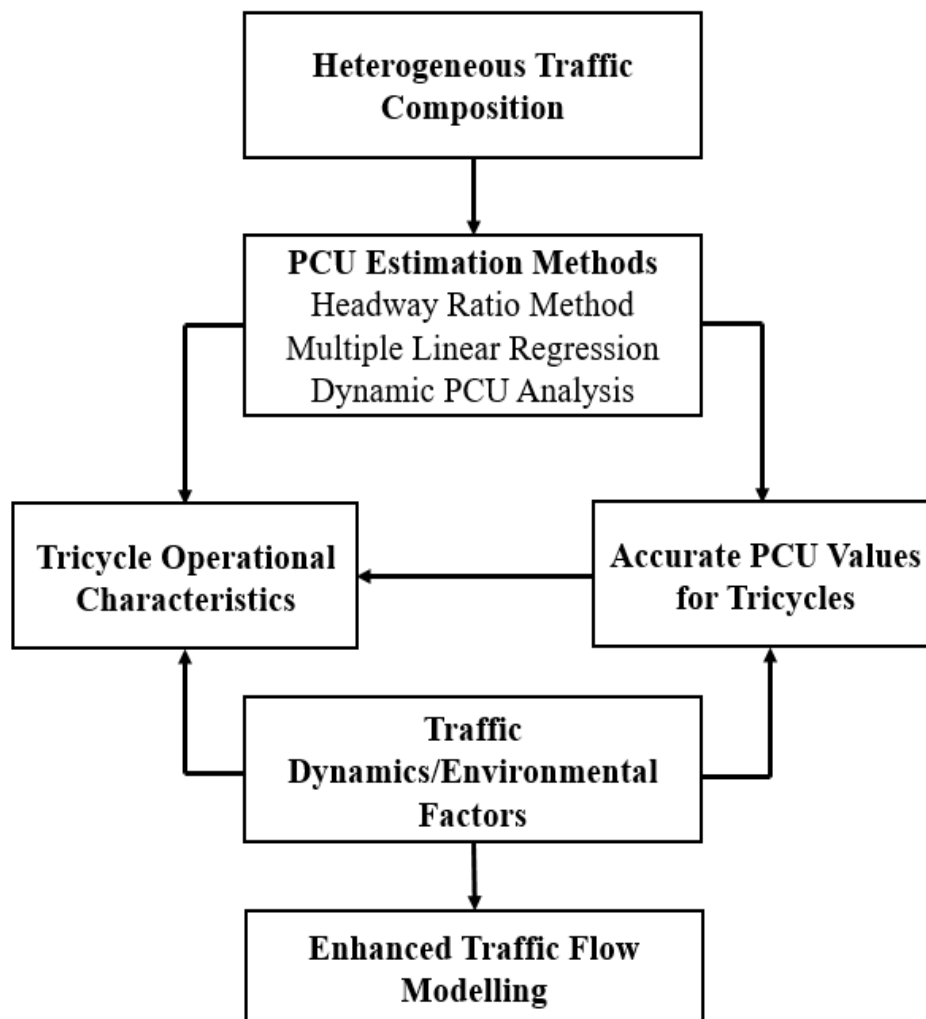


Figure 2 Conceptual Framework

CHAPTER TWO

2 LITERATURE REVIEW

2.1 Introduction

This chapter entails existing literature on Passenger Car Unit determination, focusing on methodologies used in its estimation, factors affecting PCU values, and previous studies on tricycle traffic behavior.

Current literature reveals three critical gaps that this study addresses:

- i. Lack of Behavioral PCUs: No studies currently quantify the PCU effect of 'aggressive weaving' unique to Kampala tricycles.
- ii. Temporal Granularity: Existing PCU estimates are static and do not account for the severe congestion window (18:30–20:00) identified by JICA (2022).
- iii. Simulation Parameters: There are no validated calibration parameters (e.g., standstill distances) for tricycles in VISSIM models for Uganda."

2.2 Conceptual Review

A central concept in traffic engineering is the PCU, a metric used to standardize the diverse effects of various vehicle types on traffic flow and road capacity (Rahman et al., 2019). In heterogeneous urban environments like Kampala, where the traffic stream comprises a mix of motorcycles, tricycles, passenger cars, minibuses, and non-motorized vehicles, assigning accurate PCU values is crucial for quantifying each vehicle type's impact and for effective road design, traffic management, and policy formulation (Baertsch, 2020; Olawale et al., 2017).

PCU values are determined by evaluating how a vehicle interacts with the traffic stream, considering factors such as maneuverability, size, speed, acceleration, and deceleration (Olawale et al., 2017). Larger vehicles, such as buses and trucks, typically have higher PCU values due to their greater road space occupancy and slower speeds, which disrupt traffic flow more significantly. Motorcycles, by contrast, have lower PCU values because of their small size and high maneuverability. Tricycles, however, present a unique challenge: their intermediate size, moderate speed, and distinctive movement behaviors—such as frequent stops and unpredictable lane usage—set them apart from both motorcycles and passenger cars, complicating PCU assessment (Olawale et al., 2017; Adams et al., 2014).

2.2.1 Unique Operational Characteristics of Tricycles in Kampala

Tricycles in Kampala operate under mixed traffic conditions, often lacking designated lanes and contending for space with both smaller and larger vehicles (Baertsch, 2020). Their operational behaviours such as frequent stops for passengers, lane indiscipline, and moderate speeds directly influence their effect on traffic flow and congestion, especially at intersections and during peak hours as seen in Table 2.1 (Okiza et al., 2024; Behrens et al., 2016). These behaviours distinguish tricycles from motorcycles, which are more agile, and from cars, which are faster and generally more disciplined in lane usage.

Table 2 Physical and Operational Characteristics

Feature	Tricycles (Tuk-tuks)	Motorcycles (Boda-bodas)	Passenger Cars
Typical Length (m)	2.6–3.2	1.8–2.2	3.5–4.5
Typical Width (m)	1.3–1.5	0.7–0.8	1.5–1.8
Passenger Capacity	3–4	1–2	4–5
Average Speed (km/h)	20–35	30–50	30–60
Maneuverability	Moderate	High	Moderate
Lane Discipline	Low (frequent weaving, stops)	Very low (weaving)	High
Typical Use	Short–medium urban trips	Short urban trips	Urban and peri-urban trips
Unique Behaviors	Frequent stops, unpredictable lane changes, moderate acceleration/deceleration	Weaving, high acceleration, frequent overtaking	Follows lanes, steady speed

Sources: Baertsch (2020); Okiza et al. (2024); Adams et al. (2014); Behrens et al. (2016); ChinAfrica (2024).

2.2.1.1 Why Existing PCU Models Are Insufficient for Tricycles

Most established PCU models are based on the behaviors and characteristics of conventional vehicles cars, buses, and motorcycles—often under conditions of lane discipline and relatively homogeneous traffic (Rahman et al., 2019; Chandra & Sikdar, 2020). These models do not account for the unique operational dynamics of tricycles in Kampala, such as:

- Frequent passenger stops and starts, which disrupt flow more than continuous-moving motorcycles or cars.
- Low lane discipline and frequent lateral movements, leading to unpredictable interactions with other vehicles.
- Intermediate size and speed, making their impact on congestion context-dependent and highly variable.

- iv. Variable passenger loads and operational patterns that differ from both motorcycles (lower load, higher maneuverability) and cars (higher speed, more consistent behavior).

Empirical studies in similar African and Asian cities have shown that tricycle PCU values are consistently higher than those for motorcycles but lower than for cars, and can vary significantly by location, intersection type, and traffic density (Adams et al., 2014; Raj et al., 2018; Okiza et al., 2024). For example, Adams et al. (2014) found PCU values for tricycles ranging from 0.67 to 0.75 at different intersections in Ghana, while Raj et al. (2018) observed values from 0.91 to 1.32 in Indian urban roads, underscoring the need for localized, context-specific estimation.

2.2.1.2 Implications for Kampala

Accurate PCU estimation for tricycles is essential for:

- i. Improving road capacity analysis and intersection design by reflecting the true impact of tricycles on flow and congestion.
- ii. Optimizing traffic signal timings and lane allocation to accommodate the operational realities of tricycles.
- iii. Supporting sustainable urban mobility planning, given the growing modal share of tricycles in Kampala's transport system (Baertsch, 2020; ChinAfrica, 2024).

Without Kampala-specific PCU values for tricycles, traffic models risk underestimating congestion and misallocating road space, leading to persistent inefficiencies and missed opportunities for targeted interventions (Okiza et al., 2024; Adams et al., 2014).

2.3 Theoretical Review

2.3.1 Traffic Flow and PCU Concept

The characteristics of traffic flow vary between developed and developing nations. Cars and a few other vehicles make up the majority of the traffic flow in developed nations, which is homogeneous in nature and follows a straightforward and orderly pattern. However, traffic in developing nations is known as heterogeneous, and it includes a diverse range of vehicles with various static and dynamic features. For road projects to be planned and carried out effectively, traffic flow needs to be precisely measured (Chandra and Kumar, 2023). Given the diverse traffic conditions, it is inappropriate to express traffic flow in terms of the number of vehicles that pass a specific section in a given amount of time. When the effects of each type of vehicle are expressed in terms of some common units known as Passenger Car Units (PCU) or

Passenger Car Equivalents (PCE), the difficulties associated with describing the diverse traffic volume can be resolved.

In traffic engineering, different types of vehicles share the same roadway, but their sizes, speeds, and maneuverability vary significantly. To ensure efficient traffic management and road capacity analysis, engineers use a standardized measure called the Passenger Car Unit (PCU). A Passenger Car Unit (PCU) is a measure of a vehicle's interaction with traffic compared to a standard passenger car under specific conditions (CSIR - Central Road Research Institute, 2017). PCU is a metric that expresses the impact of a particular vehicle type on traffic flow relative to a standard passenger car. Since larger or slower vehicles take up more space and time on the road compared to smaller, faster ones, their PCU values differ.

In traffic studies, the passenger car unit (PCU) is important. In the United States, the Highway Capacity Manual (HCM) coined the word "PCU" to refer to the volume or capacity of passenger automobiles per hour per lane (Lera and Kuleno, 2020). PCU was defined in the Highway Capacity Manual (Highway Research Board 1965) as "the number of passenger cars displaced in the traffic flow by a truck or a bus, under the existing roadway and traffic conditions" (Ameena and Dhurai, 2023). PCU was defined as "the measure of relative interaction between a vehicle and a traffic stream with respect to a standard passenger car under a given set of highway and traffic conditions" by the Central Road Research Institute (CSIR) in 2017. It is used as a conversion factor to compare the impact of various vehicle types, such as motorcycles, tricycles, buses, and trucks, to that of a standard passenger car in terms of traffic flow and road capacity (Lera and Kuleno, 2020). The Passenger Car Unit (PCU) is a factor that converts mixed traffic into an equivalent number of passenger cars. It is also defined as the ratio of the time taken by a given type of vehicle to traverse a road segment compared to a standard passenger car under similar conditions (Khisty and Lall, 2016). PCU values help traffic engineers assess road capacity, determine lane requirements, and optimize traffic signals for smoother flow. For example, heavy trucks and buses have higher PCU values since they occupy more space and move at lower speeds, while motorcycles and bicycles have lower PCU values due to their smaller footprint and higher maneuverability (Stewart-Wilson and Mutabazi, 2017).

PCU values are crucial for traffic analysis, capacity assessment, and roadway design. They help in estimating congestion levels, optimizing signal timings, and designing road infrastructure. The concept of PCU is crucial in traffic engineering as it facilitates traffic engineers in

standardizing heterogeneous traffic into homogeneous units, thus simplifying traffic modeling and simulation (Chandra and Sikdar, 2020).

PCU concept allows for the evaluation of road capacity in diverse traffic conditions. Roads are typically designed based on the expected number of vehicles they can handle, but since traffic is rarely composed of just cars, PCU values provide a realistic way to account for mixed traffic. This ensures that infrastructure planning is based on actual usage rather than theoretical assumptions (Vermeiren *et al.*, 2015). Also, PCU values play a vital role in traffic flow analysis and management. Engineers use PCU-based calculations to design and time traffic signals efficiently, ensuring that vehicles of different sizes receive appropriate green-light durations (Gwilliam, 2013). Without this adjustment, large vehicles could create unnecessary congestion, while smaller ones could be under-prioritized (Arnott *et al.*, 2015). PCU is essential for roadway design and lane allocation because by understanding how different vehicle types influence traffic movement, engineers can allocate lanes more effectively (Banskota and Shahi, 2021). For instance, dedicated bus lanes or truck lanes can be planned based on the PCU values of these vehicles, ensuring smoother movement and reduced congestion. Additionally, PCU assists in traffic simulation and modeling, which helps researchers and policymakers predict traffic behavior under different scenarios. By incorporating PCU values into simulations, planners can test road designs, assess the impact of increased vehicle numbers, and develop better traffic management strategies (Baertsch, 2020).

2.3.2 Methods of PCU Estimation

In order to manage traffic regulation and control, as well as to construct and evaluate the operation of various traffic facilities, it is essential to estimate PCU for each vehicle type in the traffic stream. Accurately measuring the homogenous traffic volume in mixed traffic scenarios is made possible by using the proper PCU for each vehicle category. This is crucial for delivering the required level of service. Furthermore, a situation of unexpected congestion could arise from a poor implementation of the PCU. The PCU of distinct vehicle categories can be estimated using a variety of techniques (Pooja et al., 2019).

The headway method, homogenous coefficient method, walker's method, multiple linear regression method, simultaneous equations method, speed based method, huber method, and simulation technique are among the most popular methods for estimating PCU of a vehicle category (Biswas & Sharma, 2021).

2.3.2.1 The Headway and Modified Headway Method

The headway method estimates PCU by comparing the time gap between successive vehicles as they pass a point on a roadway (Khisty and Lall, 2016). In this approach, the average headway (time interval between vehicles) of a specific vehicle type is measured and compared to that of a passenger car. A larger headway value indicates that a vehicle requires more road space and time, leading to a higher PCU value. This approach is easy to use and performs best at higher service levels and with less mixed traffic. However, in crowded or extremely dynamic traffic scenarios when vehicle interactions vary greatly, its accuracy can be jeopardized. Accurate field measurements of progress are difficult to get without the use of advanced instruments or methods. Obtaining sufficient samples from more widely dispersed automobiles is difficult. This often limits the application of the method in the field. When lane discipline is respected, the headway approach can be used (Ameena & Dhurai, 2023). As a result, the PCU model for a truck can be found in Eq. (1).

$$PCU_{Truck} = \frac{[(\frac{h_{mix}}{h_c}) - P_c]}{P_T} \quad (1)$$

Where, h_{mix} = normal time headway between two vehicles in mixed traffic stream (s)

h_c = normal time headway between two passenger cars (s)

P_c = fraction of passenger cars in the mixed traffic stream

P_T = fraction of trucks in the mixed traffic stream

Nevertheless, this approach can only be used when there are cars and another type of vehicle in the traffic stream. A modified model is provided as illustrated in Eq. (2) in order to get around this restriction. Any type of vehicle in the traffic stream can have its PCU found using this procedure.

$$PCU_i = \frac{h_i}{h_c} \quad (2)$$

Where, PCU_i = PCU of vehicle classification 'i'

h_i = normal time headway between two vehicle of type 'i' (s)

h_c = normal time headway between two passenger cars (s)

The modified headway method is a straightforward and more efficient methodology that takes into account the vehicle's speed, estimated area, and time headway (McCormick, 2013). It is

an improved version of the traditional headway method. Due to the limitations of the headway method particularly in heterogeneous traffic conditions where vehicle speeds, sizes, and acceleration patterns vary significantly, the modified headway method incorporates additional factors such as speed variability, vehicle type interactions, and lane utilization to provide more accurate PCU estimates (Lera and Kuleno, 2020). The modified headway method refines the traditional approach by accounting for factors such as vehicle classification, speed differences, and the influence of larger or slower-moving vehicles on overall traffic flow. Instead of relying solely on time gaps between vehicles, this method integrates multiple parameters to develop a more realistic estimation of a vehicle's impact on traffic movement (McCormick, 2013). This makes it particularly useful in urban environments where traffic density is high and vehicles frequently change lanes or adjust speeds.

Key improvements over the traditional headway method (McCormick, 2013; Ahmed, 2020):

- i) **Incorporation of Speed Factors:** The modified method includes speed variability in determining PCU values. It acknowledges that slower-moving vehicles create greater disruptions in traffic flow, leading to higher PCU values, while faster-moving vehicles contribute less to congestion.
- ii) **Consideration of Lane Utilization:** Unlike the traditional method, which assumes vehicles remain in a single lane, the modified approach takes into account lane-changing behavior. This is particularly relevant in mixed traffic scenarios where smaller vehicles, such as motorcycles and auto-rickshaws, weave between lanes.
- iii) **Adjustment for Vehicle Interaction Effects:** The method accounts for the influence of different vehicle types on each other's movement. For instance, heavy trucks and buses may force surrounding vehicles to slow down, thereby affecting the overall traffic flow.
- iv) **Dynamic Headway Measurement:** Instead of using a fixed time interval for headway calculation, the modified method dynamically adjusts for variations in traffic density and congestion levels, providing more adaptable PCU values.

The modified headway method is widely used in studies where accurate PCU estimation is necessary for better traffic modeling and infrastructure planning (Ahmed, 2020). The process begins with data collection, where traffic flow data, including headway times, vehicle speeds, and lane utilization patterns, are recorded using sensors, video analysis, or manual observation. Once the data is collected, vehicles are classified based on their size, maneuverability, and speed characteristics to ensure a more accurate estimation of their impact on traffic flow.

After classification, the calculation of dynamic headway ratios is performed. This involves comparing the time taken by a particular vehicle type to traverse a road section with that of a

passenger car, while adjusting for factors such as speed, lane changes, and congestion levels. Finally, the PCU values for each vehicle category are derived using modified equations that incorporate these additional parameters, ensuring a more precise estimation.

One of the key advantages of the modified headway method is its higher accuracy in mixed traffic conditions. By including multiple factors such as speed variations and vehicle interactions, this method results in a more precise estimation of PCU values (Ahmed, 2020). Another significant advantage is its ability to better represent real-world traffic behavior. Since the method takes into account lane utilization and speed variations, it provides a more practical analysis of how different vehicle types impact road capacity (Pooja *et al.*, 2019). In addition, the modified headway method improves infrastructure planning by providing reliable PCU estimates. Traffic engineers can use these estimates to design roadways, intersections, and signal systems that better accommodate actual traffic conditions, leading to more efficient transportation networks (Ogundipe, 2018). Furthermore, the method's adaptability makes it suitable for different traffic environments. It can be applied in both urban and rural settings, making it a versatile tool for traffic analysis and management.

Despite its advantages, the modified headway method also has certain challenges (Ogundipe, 2018; Ahmed, 2020; Pooja *et al.*, 2019):

- i) One major challenge is the complexity in data collection. The method requires extensive traffic monitoring and real-time data analysis, which can make its implementation resource-intensive.
- ii) Variability in traffic conditions can affect the accuracy of PCU estimation. Differences in road design, driver behavior, and vehicle characteristics may require location-specific adjustments to ensure reliable results.
- iii) Another limitation is the computational demand associated with this method. Since the modified headway method involves advanced data processing and statistical modeling, it may pose challenges in regions with limited technical resources. The requirement for specialized software and expertise in traffic data analysis can also make the application of this method more demanding.

2.3.2.2 Homogenous Coefficient Method

The homogeneous coefficient method determines PCU by observing traffic flow in homogeneous conditions where only one vehicle category is present at a time (Adnan, 2024). The volume of vehicles per unit time is recorded, and the equivalency is calculated based on the time required for the same volume of passenger cars to pass the same point. This method is

useful in controlled environments but may not always reflect real-world mixed traffic conditions where vehicle interactions play a critical role. In a mixed traffic stream, the Permanent International Association of Road Congress (PIARC) proposed a model to determine the homogeneity coefficient (or PCU) of a vehicle type. Eq. (3) illustrates how the length and speed of a vehicle were taken into consideration when developing the homogeneous coefficient.

$$HC_i = \frac{L_i/V_i}{L_c/V_c} \quad (3)$$

Where, L_i = size of vehicle 'i' (m)

V_i = normal speed of vehicle 'i' (km/h)

L_c = size of passenger car (m)

V_c = normal speed of passenger car (km/h)

2.3.2.3 Walker Method

Walker's method is based on the concept of vehicle spacing and speed variation. It considers the influence of vehicle speed, space occupancy, and headway on PCU estimation. This method is particularly effective in highways and free-flow conditions, where speed differences between vehicle categories significantly affect traffic movement. It is based on how many overtakes are made on a given road segment, presuming that each car keeps traveling at its typical speed. One of its key advantages is that it accounts for dynamic traffic behavior, making it more adaptable than static headway-based approaches. However, it requires extensive field data collection and can be complex to implement. The ratio of overtaking (N_2) when trucks are present to overtaking (N_1) when there is an equal amount of traffic composed entirely of passenger cars is known as the PCU of a slow-moving vehicle (truck). If the number of trucks in mixed traffic conditions is Q_2 , and the number of passenger cars in all-car conditions is Q_1 , then

$$PCU_{Truck} = \frac{N_2 \times Q_1}{N_1 \times Q_2} \quad (4)$$

On the other hand, it is challenging to quantify the number of overtaking (N_1 and N_2) on the field. Thus, the assessment of the number of overtaking is likewise covered by the method. If there are y vehicles per hour traveling at speed V_1 and x vehicles per hour traveling at speed V_2 within one kilometer of the road, then the number of overtaking vehicles is defined as

$$N = \sum_{i=1}^n \sum_{j=1}^m x_i y_j \left[\frac{1}{V_{2i}} - \frac{1}{V_{1j}} \right] \quad (5)$$

2.3.2.4 Multiple Linear Regression Method

The multiple linear regression approach estimates PCU values based on a number of traffic characteristics, including lane occupancy, vehicle speed, acceleration, and deceleration, using statistical analysis. This approach establishes the connection between various vehicle types and their effects on traffic flow by gathering copious amounts of traffic data and using regression models. Although it necessitates sophisticated data analysis abilities and computing tools, it is quite precise in a variety of traffic situations. Under conditions of heterogeneous traffic flow, the equivalency units of distinct vehicle types can be precisely estimated using the special multiple linear approach regression method. Under varied traffic flow situations, the PCU values calculated with this approach are shown to be reasonable and realistic (Adnan, 2014). Eq. (6) below uses a multiple linear equation to express the average speed of passenger cars based on classified traffic volumes.

$$V_C = a_0 + \sum_{i=1}^n a_i Q_i \quad (6)$$

Where, V_C = normal speed of passenger cars (km/h)

Q_i = volume of vehicle category 'i' (veh/h)

$i = 1$; for passenger cars

a_0 = intercept

a_i = regression constants

n = quantity of vehicle categories

Coefficients are expected to be negative since speed in general, reduces with the increase in traffic volume. a_0 and a_i ; $i = 1(1)n$ are determined exercising regression technique. Magnitude of a_i signifies the contribution of 'i' category vehicle in reducing stream speed. The higher value of a_i denotes higher weightage of 'i' category vehicle in the traffic stream. Hence, PCU of a vehicle category 'i' (PCU_i) is estimated using Eq. 7 below.

Since speed generally decreases as traffic volume increases, coefficients a_i should be negative. Regression analysis is used to determine a_0 and a_i ; $i = 1(1)n$. The "i" category vehicle's contribution to decreasing stream speed is shown by the magnitude of a_i . Vehicles in the "i" category are given more weight in the traffic stream when the value of a_i is higher. Therefore, Eq. 7 below is used to estimate the PCU of a vehicle category "i" (PCU_i).

$$PCU_i = \frac{a_i}{a_1} \quad (7)$$

a_1 is the weightage factor for traditional vehicle.

2.3.2.5 Speed-Based Method

By contrasting the speeds of various vehicle types with those of a typical passenger car, the speed-based approach calculates PCU. The fundamental premise is that because a slower-moving vehicle has a bigger impact on traffic flow than a passenger car, it will have a higher PCU value. This approach works well for capturing the dynamic nature of the PCU and is easy to utilize in both urban and rural traffic investigations. Additionally, it works well with diverse traffic streams that include a range of vehicle kinds. It might not, however, adequately account for the impact of elements like vehicle maneuverability and acceleration characteristics. Focusing on the influence area instead of the anticipated rectangle region would be more precise and appropriate (Pooja et al., 2018). When compared to a typical vehicle, this method indicates that PCU is inversely proportional to projected area ratio and directly proportional to speed ratio, as shown in Eq. 8 below.

$$PCU_i = \frac{V_c/V_i}{A_c/A_i} \quad (8)$$

Where, V_i and V_c = mean speeds of vehicle type 'i' and car respectively;

A_i and A_c = respective projected rectangular areas of vehicle type 'i' and car

2.3.2.6 Simultaneous Equation Method

The simultaneous equations method involves developing mathematical equations that relate various traffic flow characteristics, such as vehicle density, volume, and speed, to determine PCU values. These equations are solved simultaneously to derive PCU estimates for different vehicle categories. This method is particularly useful for understanding complex traffic interactions but requires a strong mathematical foundation and detailed data collection. It is headway-based method for estimating PCU. Simultaneous equation method is more general because it takes into account all vehicle categories. If h_1 is the average headway for an all-car scenario and h_{mix} is that for a mixed traffic stream, the volume of all-car traffic (Q_1) and of mixed traffic Q_m are calculated as follows.

$$Q_1 = \frac{3600}{h_1} \quad (9)$$

$$Q_m = \frac{3600}{h_{mix}} \quad (10)$$

Equations 9 and 10 implies that;

$$Q_1 h_1 = Q_m h_{mix} \quad (11)$$

A further definition of the mixed traffic stream's average headway is:

$$h_{mix} = h_1 \sum_{i=1}^n (P_i \times PCU_i) \quad (12)$$

P_i = fraction of vehicle category 'i'

PCU_i = PCU of vehicle classification 'i'

n = quantity of vehicle classifications

Therefore;

$$\frac{h_{mix}}{h_1} = \frac{Q_1}{Q_m} = \sum_{i=1}^n (P_i \times PCU_i) \quad (13)$$

$$Q_1 = Q_m \sum_{i=1}^n (P_i \times PCU_i) \quad (14)$$

Several simultaneous equations for comparable traffic situations can be created by applying Eq. 13 (Fisher and Hallet, 2018). Any of these methods can be used to determine the PCU of various vehicle types.

2.3.2.7 Huber Method

The Huber method is an advanced approach that considers multiple traffic parameters, including lane occupancy, headway, and speed. It is particularly useful in estimating PCU values in mixed traffic conditions where different vehicle types interact dynamically. The method provides more accurate PCU estimates but requires substantial traffic data and computational effort. This method uses traffic flow rate to determine whether a subject vehicle category other than standard passenger cars exists or not. Huber in the year 1982 proposed this method, which only considers trucks and standard cars in the traffic stream.

$$PCU = \left(\frac{1}{P}\right) \left[\frac{q_b}{q_m} - 1\right] + 1 \quad (15)$$

PCU = unit of passenger car

P = fraction of truck in mixed traffic flow

q_b, q_m = flow rate, correspondingly, for mixed and basic traffic streams.

2.3.2.8 Simulation Technique

Over the past few decades, the computer-aided simulation technique has become more and more popular (Webster and Elefteriadou, 1999; Arasan and Arkatkar, 2010; Mehar et al., 2014). A computer model that mimics the behavior of vehicles seen in the field is created using the simulation technique. The simulation technique uses traffic simulation software to model real-world traffic scenarios and estimate PCU values, therefore it is possible to predict the traffic parameters against any traffic condition using the developed model. By simulating different traffic conditions and vehicle interactions, this method provides highly accurate PCU values that reflect actual roadway behavior. It is one of the most effective techniques for PCU estimation, especially in complex traffic environments. However, it requires access to sophisticated traffic simulation tools and extensive calibration to ensure accuracy.

A vehicle category's PCU can vary depending on what method is used to derive it, so choosing the right approach is critical. The suitability of an approach is determined by the study's requirements and scope, as well as the context of the traffic in question. Highlighted below in Table 3 is a summary of the advantages and disadvantages of this various methods.

Table 3 Summary of the Advantages and Disadvantages of Various PCU Estimation Methods

Reference	Method	Advantages	Disadvantages
Alhassan and Ben-Edigbe, 2012; Sarraj and Jadili, 2012	Headway	The methodology is straightforward. It works best at higher service levels and in less varied traffic situations.	It is challenging to estimate headway accurately without the use of sophisticated tools or techniques. It is challenging to collect enough samples from automobiles that are less common. This frequently restricts the method's field application. The headway method is applicable in situations where lane discipline is observed.
Yeung et al., 2015	Homogeneous Coefficient	It is simple to measure the parameters involved.	Measuring the average speed of cars does not reveal how it varies depending on the traffic conditions. The vehicle's length alone is insufficient to evaluate how it interacts with other cars in mixed traffic.
Cunagin and Messer, 1982	Walker's	It takes into account the "number of overtaking" when evaluating PCU. In order to avoid its laborious measurement, this method also includes the estimation of the number of overtaking.	It makes the assumption that every car keeps traveling at its typical speed. When there are more than two types of vehicles in the traffic stream, this approach is not appropriate.
Adnan, 2014	Multiple Linear Regression	It works well with the extremely varied traffic.	Regression coefficients can occasionally turn out to be negative, which results in an imprecise PCU estimate.
Dhamaniya and Chandra, 2016; Biswas et al., 2017a,b	Simultaneous Equations	Compared to the multiple linear regression approach, the PCU computed using this method is more accurate.	Because of the iteration procedure, the computations are time-consuming and tiresome.
Biswas et al., 2017a,b; Biswas et al., 2020; Raj et al., 2018	Speed Based	This approach is easy to use and can represent the dynamic nature of PCUs. Additionally, it works well with mixed traffic streams that include a variety of vehicle types.	It would make more sense and be more accurate to consider "influence area" rather than "projected rectangular area."

Reference	Method	Advantages	Disadvantages
Pajecki <i>et al.</i> , 2019	Huber	This method's primary benefit is its ease of use. The traffic flow rate is used to estimate PCU as it can be measured easily in the field.	Only when one vehicle category other than the typical passenger car is present in the traffic stream can this strategy be used.
Yeung <i>et al.</i> , 2015	Simulation Technique	Because traffic and geometric circumstances may be properly controlled, it is helpful in any kind of traffic situation.	A thorough understanding of the traffic simulation is necessary. Furthermore, this model is challenging to evaluate since it necessitates real-world data across a broad range of traffic and geometry situations.

Table 4: PCU Method Comparison for Kampala's Tricycle Traffic

Method	Data Requirements	Strengths	Weaknesses	Suitability for Kampala
Headway Ratio	Time headways, lane discipline	Simple, low cost	Assumes lane discipline; ignores lateral interactions	Low
Modified Headway	Speed, lane changes, congestion	Captures dynamic interactions	Requires video/sensor data	High
MLR	Traffic volume, speed, vehicle mix	Flexible for mixed traffic	Needs statistical expertise	High
Speed-Area	Speed, vehicle dimensions	Low-cost, aligns with IRC guidelines	Ignores stopping behavior	Moderate
Simulation	Traffic flow, geometric data	Models complex interactions	Software/calibration costs	High

Therefore, the following are the prioritized methods for Kampala;

- i. Multiple Linear Regression (MLR): This is because this method balances accuracy and feasibility. Can integrate Kampala specific factors (e.g., tricycle stops, boda-boda interactions) while using manually collected traffic counts.
- ii. Modified Headway Method: This method is also suitable since it captures lane-switching and congestion effects at key sites like Bakuli Intersection. Video data can be collected via partnerships with KCCA's traffic monitoring units.
- iii. Simulation Technique: This method is also suitable since it critical for testing policy interventions (e.g., tricycle-exclusive lanes). Calibration will use empirical data from MLR and Modified Headway results.

2.3.3 Estimated Value of PCU

In traffic engineering, the PCU is a crucial metric used to express the impact of different vehicle types on traffic flow in relation to a standard passenger car. Since vehicles vary in size, speed, and maneuverability, PCU values help standardize their effect on road capacity and congestion levels. The estimated value of PCU can be determined using two primary approaches: Static PCU values and Dynamic PCU values. Each approach provides different insights into how various vehicle types influence traffic movement.

Static PCU values assign a fixed equivalency factor to each vehicle type, representing its impact on traffic flow relative to a single passenger car under uniform, controlled conditions. These values, such as those in the Highway Capacity Manual or Indonesian Highway Capacity Manual, are based on average vehicle dimensions and operational characteristics, assuming steady traffic with minimal variation. While static PCU values are straightforward and useful for infrastructure planning in consistent environments, they do not account for real-world variability in traffic, potentially leading to inaccuracies in capacity assessments when actual conditions differ from the assumed standards.

Dynamic PCU values, in contrast, reflect the fluctuating impact of different vehicle types under varying traffic conditions, taking into account factors like speed, congestion, lane discipline, and road geometry. This method uses empirical data, simulations, or regression models to estimate how vehicles interact in heterogeneous and unpredictable traffic, providing more accurate and adaptive measures for traffic management and control. However, dynamic PCU estimation is complex and data-intensive, requiring advanced monitoring and analysis to capture the influence of numerous variables, including human behavior and roadside frictions, making it more resource-demanding than the static approach.

This study prioritizes dynamic PCU estimation for tricycles in Kampala, Uganda, to account for the city's highly variable traffic conditions, characterized by:

- i. Unpredictable congestion patterns: Peak-hour traffic density fluctuates by 40–70% on major corridors like Kampala Road and Jinja Road (Okiza et al., 2024; KCCA, 2023).
- ii. Heterogeneous interactions: Tricycles exhibit lane indiscipline, frequent stops, and variable speeds (20–35 km/h) that disrupt flow unpredictably (Baertsch, 2020; ChinAfrica, 2024).
- iii. Context-dependent impacts: Tricycle PCU values vary significantly by intersection type (e.g., 0.91 at signalized junctions vs. 1.32 at roundabouts in similar cities) (Raj et al., 2018; Adams et al., 2014).

2.3.3.1 Justification for Dynamic PCU Prioritization

Traffic Variability: Kampala's traffic operates without lane discipline, and tricycles adapt their behavior to avoid congestion, rendering static PCU values inadequate (Biswas & Sharma, 2021; Dhamaniya & Chandra, 2016).

Operational Realities: Tricycles’ acceleration/deceleration rates (1.5–2.5 m/s²) and frequent stops increase their impact during peak hours, which static PCU models fail to capture (Alecsandru et al., 2023).

Policy Relevance: Dynamic PCU aligns with Kampala Capital City Authority’s (KCCA) goal of adaptive traffic management, enabling time-sensitive interventions like peak-hour lane allocations (KCCA, 2022).

2.3.3.2 Integration of Static PCU

While dynamic PCU is prioritized, **static PCU values** will be calculated using the speed-area ratio method (Eq. 8) to:

- i. Provide baseline comparisons with IRC guidelines and regional studies (e.g., Nigeria’s static tricycle PCU = 1.8–2.0) (Olawole & Olakunle, 2022).
- ii. Support infrastructure planning for non-peak scenarios (e.g., rural-urban corridors with stable traffic).

2.3.3.3 Reconciliation of Results

Dynamic and static PCU values will be reconciled through:

- i. **Statistical Weighting:** Assigning higher weights to dynamic PCU during peak hours (70%) and static PCU during off-peak (30%) based on traffic density thresholds (>50 vehicles/km = dynamic priority) (Okiza et al., 2024).
- ii. **Scenario-Based Analysis:** Presenting PCU ranges for key use cases:

Table 5: PCU ranges for key use cases

Scenario	PCU Range (Tricycles)	Method Priority
Peak-hour congestion	1.2-1.6	Dynamic (Headway/MLR)
Off-peak arterial roads	0.8-1.1	Static (Speed-Area)
Signalized intersections	1.0-1.4	Dynamic (Simulation)

- iii. **Validation:** Cross-verifying results using VISSIM simulations calibrated with field data from Wandegaya and Kibuye junctions (KCCA, 2023).

2.4 Tricycle in Urban Traffic

Urban centers in developing countries are experiencing rapid increases in traffic volumes, leading to significant congestion, economic losses, and a decline in residents’ quality of life

(Arnott et al., 2015). Kampala, Uganda, exemplifies these challenges, with the daily cost of congestion estimated at USD 1.5 million, underscoring the urgent need for effective traffic management and inclusive mobility solutions (Baertsch, 2020; Okiza et al., 2024).

2.4.1 The Role and Characteristics of Tricycles in Kampala

Tricycles, commonly known as tuk-tuks, are three-wheeled vehicles with two wheels at the back and one at the front, typically carrying three to four passengers (Oxford Press, 2017; ChinAfrica, 2024). In Kampala, tricycles have become a vital component of the urban transport system, particularly for first- and last-mile connectivity in areas underserved by formal public transport. Their affordability, ability to navigate congested and narrow streets, and flexible routing make them a preferred alternative for many commuters (Adebayo & Aluko, 2020; Behrens et al., 2016).

Recent reports indicate that thousands of tricycles operate daily in Kampala, serving both passenger and cargo needs, with typical trip lengths ranging from 3 to 10 kilometers within the city (ChinAfrica, 2024). Tricycles are especially prevalent in densely populated neighborhoods and peri-urban areas, where they provide essential access to markets, schools, and workplaces. Their operational model is characterized by frequent stops, flexible routes, and informal scheduling, which, while beneficial for users, can contribute to traffic flow disruptions and congestion (Okiza et al., 2024).

2.4.1.1 Kampala-Specific Data and Operational Challenges

- i. **Ridership:** While precise figures are unavailable, anecdotal evidence and local authority estimates suggest that tricycles account for a significant share of informal public transport trips, particularly in areas lacking minibuses or bus coverage (ChinAfrica, 2024).
- ii. **Trip Lengths:** Most tricycle journeys in Kampala are short-haul, typically under 10 km, catering to intra-city mobility and complementing longer-distance minibuses or bus routes.
- iii. **Operational Challenges:** Tricycles in Kampala face several challenges, including poor road conditions, lack of designated parking or staging areas, and limited formal regulation. Their frequent stops, unpredictable lane changes, and tendency to operate outside formal traffic rules contribute to congestion and safety risks (Okiza et al., 2024; Baertsch, 2020). Additionally, the absence of standardized fare structures and weak enforcement of licensing requirements have led to inconsistent service quality and operational inefficiencies.

2.4.2 Comparative Perspective: Kampala vs. Other African Cities

Kampala's tricycle system shares similarities with those in other African cities such as Lagos (Nigeria) and Nairobi (Kenya), where tricycles have emerged as crucial modes of transport for low-income and peri-urban populations (Nwachukwu & Onu, 2019; Behrens et al., 2016). However, several unique features and regulatory differences distinguish Kampala's context:

- i. **Regulatory Environment:** While cities like Lagos have implemented strict route restrictions and licensing regimes for tricycles, enforcement in Kampala remains limited, with tricycles operating in a largely informal environment (Ogunlana & Fakorede, 2021; Okiza et al., 2024). Kampala authorities have debated bans or restrictions, but tricycles remain widely used due to public demand and lack of alternatives (Daily Monitor, 2017).
- ii. **Integration with Formal Transport:** In contrast to some Nigerian and Kenyan cities, where tricycles are increasingly integrated as feeder services to major transit hubs, Kampala's tricycles operate with minimal coordination with formal public transport, often filling service gaps in an ad hoc manner (Olvera et al., 2016; Behrens et al., 2016).
- iii. **Socio-Economic Impact:** Tricycles provide critical employment opportunities for drivers and support staff in Kampala, mirroring trends in other African cities. They also foster small-scale entrepreneurship, with some operators using tricycles for goods delivery, water transport, or even as makeshift ambulances in rural areas (ChinAfrica, 2024).

Table 6: Kampala Tricycles vs. Other African Cities

Feature	Kampala, Uganda	Lagos, Nigeria	Nairobi, Kenya
Regulatory Environment	Limited enforcement, mostly informal	Route restrictions, licensing, periodic bans	Mixed: some restrictions, growing formalization
Typical Trip Length	3–10 km	2–8 km	2–7 km
Main Users	Low-income, peri-urban, small traders	Low-income, peri-urban, commuters	Low-income, informal sector
Integration with Formal Transit	Minimal	Increasing feeder integration	Growing integration
Operational Challenges	Poor roads, lack of staging, weak enforcement	Enforcement, periodic bans, fare instability	Congestion, fare disputes, safety concerns
Unique Features	Cargo use, water delivery, informal ambulance	Strong unions, periodic government crackdowns	Increasing digital hailing, regulatory pilots

Sources: ChinAfrica (2024); Behrens et al. (2016); Ogunlana & Fakorede (2021); Olvera et al. (2016); Okiza et al. (2024).

2.4.3 Impacts and Challenges of Tricycle

While tricycles provide significant benefits, their presence in urban traffic raises several concerns. One of the primary issues is their contribution to congestion. Due to their relatively low speed and frequent stops, tricycles can disrupt the smooth flow of traffic, particularly in high-density areas (Eze & Okafor, 2021). Furthermore, they often operate outside formal traffic regulations, leading to increased road conflicts and higher accident risks (Afolabi *et al.*, 2020). Another challenge associated with tricycles is their impact on road capacity. Studies have shown that tricycles have a Passenger Car Unit (PCU) value higher than motorcycles but lower than cars, affecting lane efficiency in mixed-traffic conditions (Olawole and Olakunle, 2022). Their unpredictable movement patterns and frequent lane switching contribute to increased travel time and reduced road efficiency (Chukwuma and Adewale, 2019).

Many urban areas struggle with enforcing traffic regulations for tricycles. In some cities, licensing and operational guidelines exist, but compliance is often poor due to weak enforcement mechanisms (Adeyemi & Ojo, 2020). For example, in Lagos, Nigeria, regulations require tricycle operators to follow designated routes, yet enforcement remains inconsistent (Ogunlana & Fakorede, 2021). The lack of standardized policies and poor rider training further exacerbate safety concerns. Some urban planners have attempted to regulate tricycle operations by restricting their movement on major roads and highways. However, such restrictions have been met with resistance, as tricycles remain an essential transport mode for many residents, particularly in low-income areas where alternative options are limited (Okonkwo *et al.*, 2022).

2.5 Gaps in Literature

Despite the increasing presence of tricycles in urban transportation, particularly in cities like Kampala, there remain significant gaps in the literature regarding their Passenger Car Unit (PCU) values.

2.5.1 Limited Research on Tricycle Traffic Behavior in Kampala

Most existing studies on PCU values focus on conventional vehicle types such as cars, motorcycles, and heavy vehicles, with little attention given to tricycles in Kampala's unique mixed-traffic environment. There is a lack of research analyzing tricycle-specific movement patterns, acceleration profiles, and lane usage in the city.

This research will collect empirical data at key intersections and corridors in Kampala (e.g., Wandegaya, Kibuye, Bakuli) to systematically analyze tricycle movement patterns,

acceleration, deceleration, and lane usage. The study will document unique operational behaviors—such as frequent stops for passengers and high maneuverability—using direct observation and video analysis.

2.5.2 Absence of Location-Specific PCU Estimations

Many PCU studies adopt generalized values from international or regional models, which do not account for Kampala’s distinctive road network, driver behavior, and traffic density. The lack of locally calibrated PCU values for tricycles results in inaccurate traffic flow predictions and suboptimal urban transport policies.

The research will generate Kampala-specific PCU values for tricycles by analyzing local traffic conditions and vehicle interactions. These values will be validated against field data and contrasted with PCU values from other cities to highlight contextual differences and improve the accuracy of traffic modeling for Kampala.

2.5.3 Inadequate Consideration of Traffic Density and Congestion Levels

Existing PCU models often assume ideal or uniform traffic conditions, failing to account for the significant fluctuations in congestion and density seen in Kampala. In reality, PCU values are dynamic and vary with road capacity, peak and off-peak hours, and congestion levels.

This study will estimate PCU values for tricycles under varying congestion scenarios by collecting data during both peak and off-peak hours, and across different seasons. The analysis will explicitly examine how tricycle PCU values change with traffic density and congestion, providing a more nuanced understanding relevant to Kampala’s conditions.

2.5.4 Limited Use of Advanced Traffic Simulation Models

Globally, advanced micro-simulation software (e.g., VISSIM, SIDRA, AIMSUN) is increasingly used to estimate PCU values, but such approaches are rarely applied to tricycles in Kampala. Reliance on traditional estimation methods may not fully capture the dynamic and heterogeneous nature of local traffic.

The research will employ both traditional (headway ratio, regression) and advanced simulation techniques to estimate tricycle PCU values. Micro-simulation tools will be calibrated using empirical data from Kampala, allowing for a robust comparison of methods and a more comprehensive understanding of tricycle impacts.

2.5.5 Lack of Integration of Tricycles into Kampala’s Transport Policy

Despite their widespread use, tricycles are not formally integrated into Kampala’s transportation planning models, and there is minimal research on how assigning accurate PCU values could enhance infrastructure planning, lane allocation, or signal optimization.

The study will explicitly evaluate the implications of integrating tricycle PCU values into Kampala’s transport policy and planning frameworks. It will provide actionable recommendations for policymakers and urban planners on road design, intersection management, and traffic signal settings, supported by empirical data.

2.5.6 Need for Empirical Data Collection and Validation

Most studies estimating PCU values rely on theoretical assumptions or data from other contexts, rather than direct field observations in Kampala.

This research will conduct comprehensive empirical data collection—including video analysis, manual counts, and automatic sensors—at representative sites in Kampala. The findings will be validated using statistical techniques and compared with regional and international benchmarks, ensuring reliability and local relevance.

2.6 Empirical Review

Table 7 below shows the summary of findings from related literatures.

Table 7 Summary of findings from the Literature Review

Author and Year	Topic	Aim	Methodology	Findings	Gaps Identified
Kumar (2011)	Evolution of Paratransit Systems in Sub-Saharan Africa	To analyze the development and challenges of paratransit systems, including minibuses and tricycles, in Sub-Saharan African cities.	Literature review and case studies	Identified the dominance of informal transport modes due to the collapse of formal public transport systems.	Lacked specific data on tricycle operations and their impact on traffic flow.
Jean et al. (2018)	Growth Trends in Kampala's Minibus Taxi Fleet	To analyze the growth rate of minibus taxis in Kampala.	Statistical analysis of vehicle registration data	The minibus taxi fleet is growing at an annual rate of 5.4%.	Did not consider the growth and impact of tricycle usage.
Aggrey (2017)	Commuter Preferences in Kampala	To assess commuter preferences for different transport modes in Kampala.	Surveys and focus group discussions	82.6% of commuters use minibus taxis; the rest use private cars, buses, and motorcycle taxis.	Did not explore the role of tricycles in commuter preferences.

Author and Year	Topic	Aim	Methodology	Findings	Gaps Identified
Stewart-Wilson et al. (2017)	Ownership Structures in Kampala's Paratransit Industry	To investigate the ownership patterns within Kampala's paratransit system.	Surveys and interviews	Ownership is diverse, ranging from individual owners to political figures.	Did not link ownership structures to traffic performance metrics.
Adams et al. (2014)	PCU Values for Urban Mixed Traffic Flow	To estimate PCU values for motorcycles, tricycles, cars, and buses/trucks at urban intersections in Ghana.	Video-based manual counts and regression analysis	Found tricycle PCU values in Ghana between 0.67 and 0.75 at signalized intersections.	No Kampala-specific PCU values; context differences.
Okiza et al. (2024)	Traffic Flow Modeling in Kampala	To model traffic flow and estimate PCU values in Kampala's mixed traffic.	Moving observer method and empirical data	Highlighted the lack of empirical PCU data for tricycles in Kampala and the need for context-specific values.	Did not provide validated PCU values for tricycles; called for empirical research.
ChinAfrica (2024)	Socio-economic Impact of Tricycles in Kampala	To document the uses and impacts of tricycles in Kampala's transport system.	Field interviews and case studies	Tricycles are widely used for passenger and goods transport, providing jobs and affordable mobility.	No quantitative analysis of traffic flow or PCU impacts.

Author and Year	Topic	Aim	Methodology	Findings	Gaps Identified
KCCA (2023)	Road Safety and Traffic Congestion in Kampala	To analyze traffic safety and congestion trends in Kampala.	Analysis of police crash records and congestion data	Identified congestion hotspots and rising crash rates, especially involving motorcycles and informal modes.	Lacks disaggregated data on tricycle involvement and PCU contribution.
Ndibatya & Booyesen (2021)	Minibus Taxi Movement Patterns in Kampala	To characterize the movement patterns of minibus taxis in Kampala.	GPS-based floating car data	Revealed self-organizing, demand-responsive patterns in paratransit.	Focused on minibus taxis; tricycles not included in movement analysis.

The literature review reveals a persistent scarcity of Kampala-specific, empirical research on tricycle PCU values and their operational impacts. Most studies either focus on minibus taxis, motorcycles, or general paratransit trends, with limited quantitative analysis of tricycle movement, lane usage, and congestion effects in the Kampala context (Adams et al., 2014; Okiza et al., 2024; ChinAfrica, 2024; KCCA, 2023).

This study will fill these gaps by collecting empirical traffic data at selected high-tricycle-density intersections and corridors in Kampala, using both video analysis and automated counting to capture unique tricycle movement patterns, lane discipline, and operational characteristics. It will also estimate and validate location-specific PCU values for tricycles using multiple established and advanced methods (headway ratio, multiple linear regression, dynamic estimation, and simulation), tailored to Kampala's mixed and informal traffic environment. The study will further analyse tricycle impacts under different congestion scenarios, including peak and off-peak periods, and comparing results with findings from other African cities to highlight context-specific differences. It will also engage with local stakeholders, such as KCCA and tricycle associations, to ensure relevance, accuracy, and policy uptake of the results. The study shall then provide actionable recommendations for integrating tricycles into Kampala's traffic models and urban transport policy, supporting more effective road design, lane allocation, and congestion management.

2.7 Implications for Policy and Practice in Kampala

Drawing on the reviewed evidence, this subsection outlines potential policy and design implications for Kampala's tricycle system.

2.7.1 Strategies for Effective Integration of Tricycles into Urban Transport Systems

To maximize the benefits of tricycles while mitigating their negative impacts, urban transport authorities must adopt context-sensitive strategies for their effective integration, especially in cities like Kampala where infrastructure and regulatory challenges are distinct.

2.7.1.1 Designation of Exclusive Lanes for Tricycles

For Kampala, preliminary feasibility checks indicate that citywide implementation of dedicated tricycle lanes is not currently realistic due to narrow right-of-way on many arterials, competing demand for NMT space, and significant on-street trading activities (KCCA, 2023; JICA, 2010). Candidate pilot corridors will therefore be restricted to sections where:

- (i) Effective carriageway width after accounting for NMT and parking is at least 10.5 m;

- (ii) No major road rehabilitation works are planned during the study period; and
- (iii) Enforcement presence by Kampala Metropolitan Police can be ensured to maintain lane discipline. These criteria will guide the selection of any pilot corridors tested using the calibrated PCU and simulation results.

2.7.1.2 Improvement of the Regulatory Framework

Kampala faces persistent challenges in enforcing licensing, training, and operational guidelines for tricycle operators (Adeyemi & Ojo, 2020; Okiza et al., 2024). Strengthening the regulatory framework should include periodic road safety training, mandatory operator licensing, and targeted enforcement at congestion-prone junctions. The Kampala Capital City Authority (KCCA) has begun to pilot digital registration and operator training schemes, but coverage remains limited (KCCA, 2023). Enhanced collaboration with tricycle associations and community leaders can improve compliance and foster safer driving practices (Ekezie & Umeh, 2023).

2.7.1.3 Integration into Formal Public Transport Networks

Tricycles in Kampala currently operate largely informally, with minimal coordination with formal public transport (Okeke et al., 2022; Stewart-Wilson & Mutabazi, 2017). Urban planners should pilot the integration of tricycles as official feeder services to major taxi parks and BRT (Bus Rapid Transit) corridors, especially in peri-urban areas where minibus coverage is sparse (JICA, 2010). Such integration could involve designated pick-up/drop-off points at transport hubs and the use of digital platforms to coordinate services, as seen in pilot initiatives in Nairobi and Lagos.

In the empirical component of this study, integration will be operationalized through measurable indicators at selected junctions and hubs, including:

- (i) Average transfer walking distance between tricycle drop-off points and formal public transport boarding points;
- (ii) Average tricycle dwell time at designated and undesignated stages.
- (iii) Observed queuing and blocking events of formal vehicles by tricycles. Improvements in these indicators under hypothetical operational changes (e.g., consolidated staging points) will be tested via simulation as part of the policy implications

2.7.1.4 Investment in Modern, Environmentally Friendly Tricycle Models

Transitioning to cleaner, more efficient tricycle models can help reduce emissions and improve air quality in Kampala (Adekunle et al., 2021). Given the city's air pollution challenges and

the government's interest in sustainable urban mobility, pilot programs for electric or low-emission tricycles could be launched in partnership with local manufacturers and international donors. These pilots should be accompanied by incentives for operators to upgrade vehicles and by the development of charging/refueling infrastructure.

CHAPTER THREE

3 METHODOLOGY

3.1 Study area

The study will be conducted in Kampala City, Uganda's capital and largest urban center, which is characterized by rapid urbanization, high population density, and increasing reliance on informal transport modes such as tricycles (commonly referred to as tuk-tuks). Kampala's road network is a complex mix of arterial, collector, and local roads, frequently experiencing congestion due to a highly heterogeneous traffic stream that includes motorcycles (boda-bodas), minibuses, private cars, and non-motorized transport (KCCA, 2023; Okiza et al., 2024).

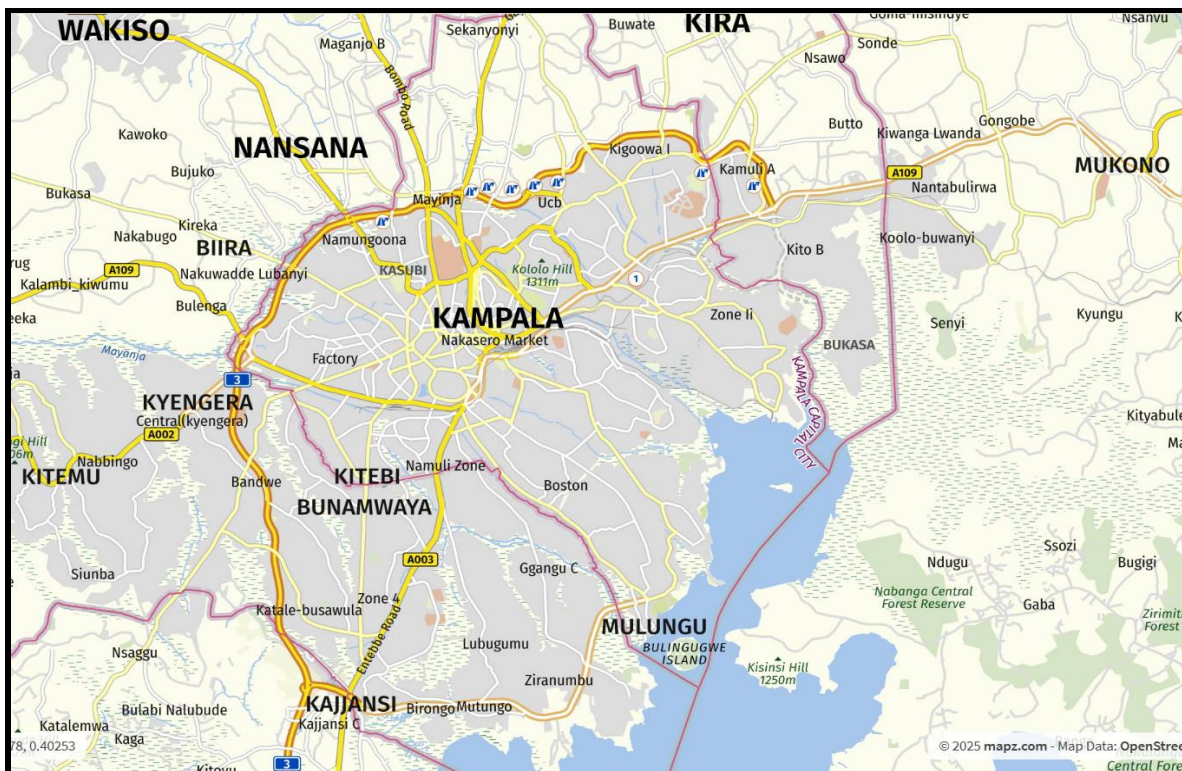


Figure 3 Study Area Map

The research will focus on key road corridors and major junctions within the Central Business District (CBD) and surrounding high-traffic areas, specifically:

- i. Wandegeya Junction: Selected due to its proximity to Makerere University, high commercial activity, and status as a major entry/exit point to the CBD. Traffic counts from JICA (2010) and KCCA (2023) consistently identify Wandegeya as a congestion hotspot, with peak hour volumes exceeding 7,000 vehicles/hour and significant tricycle activity.

- ii. Bwaise Junction: A critical northern gateway to the city, Bwaise is known for heavy traffic volumes, frequent flooding, and high modal diversity. JICA (2010) reports Bwaise as among the top intersections for vehicle throughput, with notable tricycle and boda-boda concentrations.
- iii. Bakuli Intersection: Serves as a major node connecting the CBD to western suburbs and markets. Traffic surveys highlight Bakuli as a bottleneck with high tricycle flows, especially during peak trading hours (KCCA, 2023).
- iv. Natete Junction: Functions as a southwestern entry point to Kampala, linking peri-urban areas to the city core. Natete is recognized for its high volume of informal transport and as a staging area for tricycles and minibuses (JICA, 2010).
- v. Kibuye Roundabout: Identified as one of Kampala's busiest and most congested intersections, Kibuye handles traffic from Entebbe Road, Ggaba Road, and the southern suburbs. JICA and KCCA data show persistent congestion and high tricycle density, making it ideal for observing mixed-traffic interactions.

These locations were selected based on a number of factors. They are characterized by high tricycle density and frequent interaction with other vehicle classes, as confirmed by recent traffic counts and congestion studies (JICA, 2010; KCCA, 2023). They are also known congestion hotspots, with peak hour volumes that often exceed intersection capacity (e.g., Wandegeya, Kibuye). They also have available space and infrastructure for safe placement of observation and data collection equipment. They have a varied road geometries and intersection types, enabling analysis of tricycle impacts under diverse real-world conditions.

3.2 Research Design

This study will employ a quantitative research design, combining field data collection, statistical analysis, and comparative evaluation to determine the PCU values for tricycles in Kampala City, Uganda. The research will begin with the selection of representative road segments and intersections within the city where tricycles are commonly used. Traffic data including vehicle type, speed, headway, and traffic volume, will be collected using video cameras and manual counting during peak and off-peak hours to capture a comprehensive range of traffic conditions. The data will then be categorized based on vehicle types; tricycles (classified by size), motorcycles, cars, and other relevant modes of transport. The study will apply established PCU estimation methods, including the headway ratio method, multiple linear regression analysis, and dynamic PCU estimation, to calculate PCU values for each

tricycle category. Statistical software will be used to perform regression modeling and evaluate the influence of traffic and roadway parameters on PCU values. The resulting PCU values will be compared with existing values from similar urban settings to assess contextual differences and validate the reliability of the findings. This systematic approach will ensure a robust and context-specific determination of PCU values that can improve the accuracy of traffic flow modeling and planning in Kampala City.

For the sampling strategy, data will be collected over a period of 20 consecutive days, including both weekdays and weekends, to capture temporal variations in traffic flow (Okiza et al., 2024). Each study site will be observed for a minimum of four hours per day, split between morning peak (6:30–9:30 a.m.), midday off-peak (11:00 a.m.–1:00 p.m.), and evening peak (4:00–7:00 p.m.) periods. This distribution ensures that both peak and off-peak conditions are adequately represented and allows for the analysis of diurnal traffic patterns (JICA, 2010). Key intersections and corridors Wandegeya, Bwaise, Bakuli, Natete, and Kibuye will be sampled on a rotating schedule, with at least three non-consecutive observation days at each location. This approach controls for day-of-week effects and ensures that no single location disproportionately influences the results. The selection of sites is based on documented high tricycle density and congestion (KCCA, 2023; JICA, 2010). Data collection will occur during both dry and wet seasons to account for seasonal variability in traffic behavior (Okiza et al., 2024). Observations will be paused or annotated during extraordinary events (e.g., major roadworks, public holidays, political rallies) to avoid skewing the dataset. Any such interruptions will be documented, and additional observation days scheduled as needed to maintain data integrity.

3.3 Mapping of Gaps to Methods

Table 8 below shows the mapping of the identified gaps to methods to be used.

Table 8: Mapping of Gaps to Methods

SN	Gaps	Method
1	Absence of Kampala-specific tricycle PCU values	Headway ratio method and modified headway analysis applied to field-observed headways for tricycles and cars at each site.
2	Lack of dynamic PCU estimates across congestion levels and times of day	Dynamic PCU estimation using segmented datasets by peak/off-peak, wet/dry conditions and intersection type

SN	Gaps	Method
3	Limited understanding of tricycle interactions with other modes	Multiple linear regression models with stream speed or flow as dependent variables and class-specific volumes, density and lane occupancy as predictors.
4	No calibrated microsimulation parameters for tricycles	VISSIM calibration using field data (flows, speeds, queues) from Wandegaya and Kibuye to derive tricycle car-following and lane-changing parameters

3.4 Data Collection

In the study, a detailed and multi-instrumental approach will be employed to ensure comprehensive and reliable data collection under mixed traffic conditions. High-definition video cameras (minimum 1080p resolution at 30–60 frames per second) will be mounted on existing roadside structures (e.g., lighting poles, building facades) at heights of 5.0–6.0 m to minimise occlusion from larger vehicles. Camera positioning will ensure a clear field of view covering at least 100 m of each approach and the entire intersection area. Recording will be restricted to daylight hours between 06:30 and 18:30 to support reliable manual classification. To manage occlusion, overlapping camera coverage will be used at complex junctions (e.g., Kibuye) and video segments with severe occlusion will be flagged and excluded from PCU estimation. Inter-observer reliability will be assessed by having at least 20% of the video segments independently coded by two trained enumerators. A minimum Cohen’s kappa coefficient of 0.8 will be required for acceptance; if this threshold is not met, additional training and re-coding will be undertaken.

Vehicles will be classified into tricycles (by size and operational class), motorcycles, cars, and other relevant categories. Special attention will be given to distinguishing passenger tricycles from cargo tricycles, with the latter excluded from the analysis to align with the study’s focus on urban passenger mobility. To ensure accurate differentiation of tricycles from motorcycles and small cars in Kampala’s mixed traffic, a multi-layered approach combining manual and automated methods will be employed, with rigorous quality assurance protocols.

3.4.1 Tricycle Identification Criteria

3.4.1.1 Physical Characteristics (Video Analysis):

- i. Structural Features: Tricycles in Kampala are distinctively three-wheeled, with a sidecar attached to a motorcycle base. Manual enumerators will be trained to identify this

configuration, which differs from motorcycles (two wheels) and small cars (four wheels) (ChinAfrica, 2024; KCCA, 2023).

- ii. Dimensions: Tricycles typically measure 2.6–3.2 meters in length and 1.3–1.5 meters in width, larger than motorcycles (1.8–2.2 meters) but smaller than compact cars (3.5–4.5 meters) (Okiza et al., 2024).
- iii. Color and Markings: Many Kampala tricycles feature standardized color schemes (e.g., yellow bodies with black stripes) and registration plates, aiding visual identification (KCCA, 2022).

3.4.1.2 Automated Classification:

- i. Axle Count and Length: Pneumatic tube counters and induction loops will classify vehicles based on axle count (tricycles: 3 axles; motorcycles: 2 axles) and length thresholds (tricycles: 2.6–3.2m) (Icoms Detections, 2024).
- ii. Speed Profiles: Tricycles in Kampala average 20–35 km/h, slower than motorcycles (30–50 km/h) but comparable to cars in congested conditions (Baertsch, 2020).

3.4.2 Enumerator Training and Reliability Checks

3.4.2.1 Training Program

Enumerators will undergo a 5-day training program covering:

- i. Image Recognition: Reviewing photo/video examples of Kampala-specific tricycle models (e.g., Bajaj RE, TVS King).
- ii. Field Practice: Conducting supervised counts at pilot sites (Wandegeya, Kibuye) to refine classification accuracy.
- iii. Software Use: Training on video analysis tools (e.g., Tracker Video Analysis) for frame-by-frame review.

3.4.2.2 Inter-Observer Reliability:

- i. Double-Blind Reviews: 20% of video footage will be independently analyzed by two enumerators. Consistency will be measured using Cohen’s kappa (κ), with a target $\kappa \geq 0.85$ (Science.gov, 2018).
- ii. Weekly Audits: Supervisors will cross-check 10% of manual counts against automated data, resolving discrepancies through consensus.

3.4.3 Data Quality Assurance

3.4.3.1 Cross-Validation:

- i. Triangulation: Manual counts from video analysis will be compared with automated sensor data hourly. Discrepancies $>5\%$ will trigger re-analysis.
- ii. Peak-Hour Focus: Additional enumerators will be deployed during peak periods (7–9 a.m., 4–7 p.m.) to handle high-density traffic and reduce miscounts.

3.4.3.2 Equipment Calibration:

- i. Daily Checks: Pneumatic tubes and induction loops will be calibrated each morning using known vehicle samples (e.g., pre-measured tricycles).
- ii. Weather Mitigation: Cameras will use infrared filters during rain, and data from severely adverse weather will be flagged/excluded (FHWA, 2004).

3.4.3.3 Error Handling:

- i. Ambiguous Cases: Blurry or occluded vehicles in video footage will be categorized as "unclassified" and excluded from final analysis ($<2\%$ target).
- ii. Sensor Failures: Redundant sensors will be installed at each site, with data gaps filled via linear interpolation from adjacent time intervals.

3.5 Data Processing and analysis techniques

3.5.1 Data Processing Workflow

The analysis will follow a structured workflow to ensure methodological rigor and reproducibility.

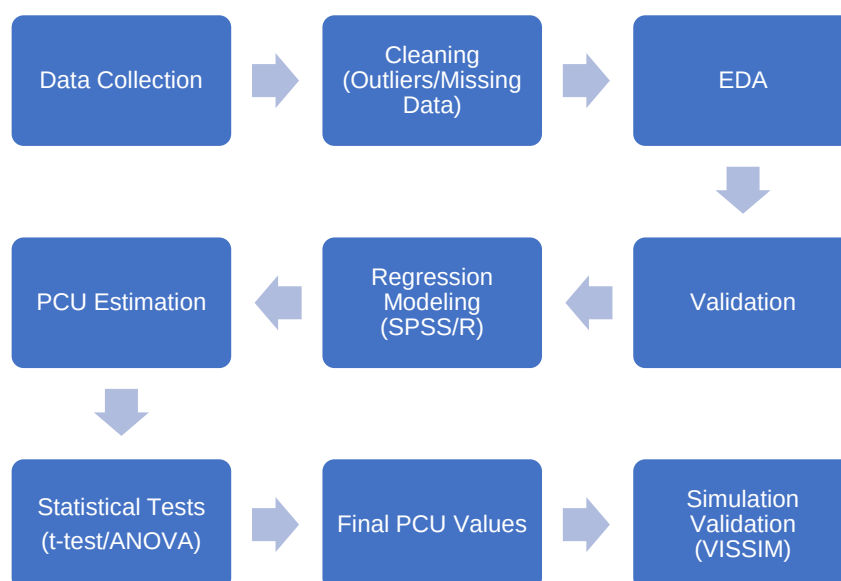


Figure 4: A flow chart showing process of data processing and analysis

i. Data Cleaning

Pre-Analysis Plan: Prior to formal analysis, a pre-analysis plan will be documented specifying:

- (i) The precise headway definition (time gap between the front bumpers of successive vehicles crossing a fixed reference line)
- (ii) The speed estimation method (time-distance measurements over known sections from video using Tracker Video Analysis)
- (iii) The thresholds for outlier detection (values exceeding 3 standard deviations from the mean traffic speed or headway will be flagged, with winsorisation applied at the 1st and 99th percentiles where appropriate)
- (iv) The criteria for variable inclusion in regression models ($VIF < 5$ and theoretical relevance). This plan will be registered with the supervisors and REC before full-scale modelling

Missing Data: Gaps will be addressed via linear interpolation for minor gaps (<5% of dataset) or probabilistic principal component analysis (PPCA) for larger gaps, leveraging traffic flow periodicity (Aman & Parti, 2021; arXiv, 2024).

Validation: Automated sensor data will cross-check manual counts; discrepancies >5% will trigger reanalysis.

ii. Exploratory Data Analysis (EDA)

Visualization: Distributions of speed, headway, and density will be plotted using boxplots and histograms to identify traffic regimes (peak vs. off-peak).

Pattern Identification: Heatmaps of tricycle lane usage and congestion hotspots will be generated for Wandegeya, Kibuye, and Bakuli intersections.

iii. Regression Modeling

Multiple Linear Regression: Applied in SPSS v28 to model PCU as a function of traffic volume, speed, density, and lane occupancy (Chandra & Sikdar, 2020). Multicollinearity will be assessed using variance inflation factors ($VIF < 5$).

Dynamic PCU Estimation: Time-series segmentation in R (v4.3.2) to model PCU variations by time of day and congestion level, with autoregressive integrated moving average (ARIMA) for peak-hour adjustments (Okiza et al., 2024).

iv. Validation

Statistical Tests:

- **One-sample t-test:** Compare Kampala's tricycle PCU values (e.g., 1.2–1.6) to benchmarks from Lagos (1.3–1.5) and Accra (0.9–1.1) at $\alpha = 0.05$ (IOSR-JMCE, 2021).
- **ANOVA:** Assess PCU differences across intersection types (signalized vs. un signalized) and vehicle classes (tricycles vs. motorcycles).

Microsimulation will be conducted using VISSIM. Key car-following (Wiedemann 74) and lane-changing parameters for tricycles (e.g., standstill distance, following variation, lane-change aggressiveness) will be tuned. Calibration quality will be assessed using: (i) the GEH statistic for link volumes, targeting $GEH < 5$ for at least 85% of links and turning movements; and (ii) Mean Absolute Percentage Error (MAPE) for average queue lengths and delays at each approach, targeting $MAPE \leq 15\%$. Only parameter sets meeting both criteria will be accepted as calibrated, and these sets will then be used to test alternative control and lane allocation scenarios.

3.5.2 PCU Estimation Methods

3.4.2.1 Headway Ratio Method

This shall address objective 1 of the study (determination of PCU values for tricycles by operational class) and Research Question 1. This method compares the average headway for tricycles ($H_{tricycle}$) to that for passenger cars (H_{car}), using the formula:

$$PCU_{tricycle} = \frac{H_{tricycle}}{H_{car}} \quad (16)$$

where $H_{tricycle}$ and H_{car} are measured in seconds.

Headways will be extracted from video data for both tricycles and cars at each observation site and time period. PCU values will be calculated separately for different traffic densities and intersection types. This method is simple and intuitive, but assumes lane discipline and may not fully capture lateral interactions in Kampala's mixed traffic.

3.4.2.2 Multiple Linear Regression Analysis

This shall address Objective 2 (comparison with other vehicle types) and Research Question 2. A multiple linear regression model will be used to quantify the impact of each vehicle type on traffic flow, using variables such as traffic volume, speed, density, and lane occupancy:

Multiple Linear Regression (MLR) models will be estimated with stream speed and/or discharge flow as dependent variables and class-specific flows as key predictors:

$$Y = \beta_0 + \beta_1 Q_{tri} + \beta_2 Q_{moto} + \beta_3 Q_{car} + \beta_4 \text{Density} + \beta_5 \text{LaneOcc} + \beta_6 \text{Rain} + \beta_7 (Q_{tri} \times \text{Density}) + \epsilon \quad (17)$$

where Q_{tri} , Q_{moto} and Q_{car} are the volumes of tricycles, motorcycles and cars (veh/h), Density is vehicles/km, LaneOcc is the proportion of time tricycles occupy more than one lane, Rain is a binary variable for wet pavement, and $Q_{tri} \times \text{Density}$ captures the interaction between tricycle flow and congestion. Multicollinearity will be assessed using variance inflation factors (VIF), with a threshold of 5.0 for acceptable values. Model performance will be evaluated using adjusted R^2 and residual diagnostics. To assess generalisability, k-fold cross-validation with $k=5$ will be conducted, and model specifications will be refined accordingly.

3.4.2.3 Dynamic PCU Estimation

This shall address objective 3 (dynamic PCU analysis under varying conditions) and Research Question 3. Dynamic PCU estimation models how PCU values for tricycles change with temporal and contextual factors:

$$PCU_{tricycle,t} = f(\text{Time of Day, Congestion Level, Intersection Type, Queue Length, Weather}) \quad (18)$$

where t denotes different time periods or scenarios.

To avoid drawing policy conclusions solely from peak-hour conditions which could unfairly portray tricycles as inherently problematic the analysis will explicitly stratify PCU estimates by time-of-day category (morning peak, off-peak, evening peak, weekend). Differences between these strata will be reported, and any regulatory or design recommendations will be based on a balanced interpretation of both average-day and peak-period performance rather than peak conditions alone

3.6 Research Ethics and Data Privacy

This study will adhere to the highest ethical standards in the collection, storage, and use of data. Video recordings and manual observations will be conducted exclusively in public spaces, with no attempt to identify individuals. All data will be anonymized, and footage will be stored securely with access limited to the research team. Where required, permissions will be sought from the KCCA and relevant local authorities. Enumerators will be trained on ethical conduct, including non-interference in traffic and respect for privacy. The research protocol will be submitted for review and clearance by an accredited institutional ethics committee prior to fieldwork.

Prior to installation of cameras, formal permission will be obtained from KCCA and local division authorities. Community leaders (LC1) at each site will be informed about the purpose, duration and nature of the video recordings, and public notices stating ‘Traffic Survey in Progress. No facial identification’ will be displayed at visible locations. Although recordings occur in public right-of-way, individuals who express concerns or objections to being filmed will be informed that only aggregate, anonymised traffic data (vehicle counts and categories) will be retained and that no attempt will be made to identify drivers or pedestrians. Contact details of the principal investigator and REC will be provided on the notices for any further queries or complaints.

To prevent third-party misuse, raw video files will be stored on encrypted drives on a secure server with access limited to the principal investigator and a designated data manager. No raw video will be shared with external agencies, including law enforcement, and no screenshots or clips showing identifiable faces or license plates will appear in reports or presentations. Only aggregated traffic indicators (e.g., classified counts, headway distributions, PCU estimates) will be disseminated. Access logs will be maintained to record every instance of file access and ensure auditability.

Raw video data will be retained for a maximum of five years after completion of the study to allow for verification and potential secondary analysis. After this period, all raw footage will be permanently deleted using secure deletion protocols, while anonymised, aggregated datasets and codebooks will be archived for longer-term research use. Access to raw video will be restricted to the principal investigator and one authorised data manager, whereas anonymised derived datasets may be shared with supervisors and collaborators under data-sharing agreements. Compliance with these retention and access rules will be monitored via annual

checks of the storage system and a simple access log which records user, date, and purpose of each access event.

3.7 Risk Assessment and Mitigation

Potential risks to project success include equipment malfunction (e.g., camera or sensor failure), adverse weather affecting data quality, observer bias, and disruptions from public events or roadworks. To mitigate these risks, the study will deploy redundant data collection equipment at each site, schedule flexible observation periods to accommodate weather or unforeseen disruptions, and implement rigorous enumerator training and cross-validation. All deviations from the data collection plan will be documented, and contingency days are built into the timeline to ensure data completeness. Regular supervisory audits will ensure data integrity and adherence to protocols.

Field safety risks will be minimised by coordinating installation times with KCCA and Kampala Metropolitan Police to avoid obstructing live traffic. Enumerators and technicians will wear high-visibility clothing and operate only from safe locations (sidewalks, medians or adjacent buildings), without standing in live lanes. Warning cones and temporary signage will be deployed where necessary during installation. No data collection will be undertaken during extreme weather or in situations where political rallies, protests or other events compromise safety; such periods will be documented and alternative days scheduled.

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APPENDICES

Appendix 1; Activity Plan and Timeline

The research is expected to be completed within 8 months, with the following breakdown:

Item No.	Activity & Milestones	Duration (Month[s])
1	Concept Preparation, Literature Review, and Proposal Development	3
	Stakeholder analysis and engagement (KCCA, tricycle unions, local communities)	
	Detailed review of Kampala-specific and regional literature	
	Drafting and revision of research proposal	
2	Data Collection and Preliminary Analysis	2
	Pilot data collection at one site to test instruments and enumerator training	
	Main data collection (video, manual, and automated counts) at all selected sites	
	Daily data upload, backup, and preliminary quality checks	
	Initial exploratory data analysis (EDA) and data cleaning	
3	In-depth Data Analysis and PCU Development	2
	Advanced statistical modeling (regression, dynamic PCU estimation)	
	Comparative analysis with benchmark cities/studies	
	Simulation and scenario testing (e.g., VISSIM calibration)	
	Drafting of interim findings and validation with field observations	

Item No.	Activity & Milestones	Duration (Month[s])
4	Validation and Refinement of the PCU Developed	1
	Stakeholder feedback sessions (KCCA, transport unions, planners)	
	Incorporation of feedback and further model refinement	
	Final validation of PCU values using statistical tests and simulation	
5	Report Writing & Submission	1
	Drafting of final report and policy briefs	
	Internal review and editing	
	Submission to university and dissemination to stakeholders	

Appendix 2; Budget

The research will require funding for various Desk and field activities. An estimated budget of UGX 20,000,000 is projected.

Budget Item	Description	Estimated Cost (UGX)	Estimated Cost (USD)
Personnel Cost	Enumerator stipends, field supervisors, and data entry staff for pilot and main data collection. Covers training, reliability checks, and overtime for peak-hour and weekend shifts.	5,000,000	1,300
Transport & Logistics	Vehicle hire, fuel, and per diem for field teams to reach all selected sites (CBD, Wandegeya, Bwaise, Bakuli, Natete, Kibuye). Includes contingency for additional trips due to weather or equipment issues.	3,500,000	900
Data Collection Equipment	Purchase/rental of high-definition video cameras (2–3 units), pneumatic tube traffic counters (2 units), GPS devices for site mapping, and spare batteries/memory cards. Includes equipment insurance and minor repair allowance.	4,000,000	1,050
Software & Data Analysis Tools	Primarily open-source software (R, Python, QGIS) for statistical and spatial analysis to minimize costs. Budget includes possible short-term SPSS license if required for advanced regression, and cloud storage for data backup.	2,000,000	525

Budget Item	Description	Estimated Cost (UGX)	Estimated Cost (USD)
Printing & Documentation	Printing of survey forms, checklists, training manuals, interim and final reports, and policy briefs. Includes binding and duplication for stakeholder dissemination.	1,500,000	400
Stakeholder Consultation	Costs for organizing meetings/workshops with KCCA, tricycle unions, and local policymakers. Covers venue hire, refreshments, and transport for key informants. Ensures engagement and feedback throughout the study.	2,500,000	650
Miscellaneous Expenses	Contingencies: Equipment repair/replacement (UGX 500,000), data backup (cloud storage, UGX 250,000), communication (mobile data, calls, UGX 300,000), ethical approval fees (UGX 250,000), administrative costs (UGX 200,000).	1,500,000	400
Total		20,000,000	5,225